11. Sequences and Series

11.9. Representations of Functions as Power Series

11.9 Learning Objectives
- Use the idea of geometric series to write functions as power series
- Find the derivative and integral of power series
- Use derivatives and integrals to find power series of more complex functions

Example 11.9.1

Find a power series representation for $\ln(1+x)$. Find its radius of convergence.

Solution

We need to relate this to something like $\frac{1}{1-u}$ in some way. Observe that $\frac{\mathrm{d}}{\mathrm{d}x}(\ln(1+x)) = \frac{1}{1+x} = \frac{1}{1-(-x)}$. So,

$$\begin{aligned}
\ln(1+x) &= \int \frac{1}{1+x}\,\mathrm{d}x \\
&= \int \frac{1}{1-(-x)}\,\mathrm{d}x \\
&= \int \left(\sum_{n=0}^{\infty}(-x)^n\right)\mathrm{d}x \\
&= C + \sum_{n=0}^{\infty}\left((-1)^n \int x^n\,\mathrm{d}x\right) \\
&= C + \sum_{n=0}^{\infty}\frac{(-1)^n x^{n+1}}{n+1}
\end{aligned}$$

Now we have to find C, using a point we know. Let $x=0$, then $\ln(1+0)=0$ and this should match the power series at $x=0$, so

$$\begin{aligned}
0 &= C + \sum_{n=0}^{\infty}\frac{(-1)^n 0^{n+1}}{n+1} \\
&= C + 0
\end{aligned}$$

So $C=0$. Thus,

$$\ln(1+x) = \sum_{n=0}^{\infty}\frac{(-1)^n x^{n+1}}{n+1}.$$

To find the radius of convergence, we can apply the ratio test

$$\begin{aligned}
L &= \lim_{n\to\infty}\left|\frac{x^{n+2}}{n+2}\cdot\frac{n+1}{x^{n+1}}\right| \\
&= \lim_{n\to\infty}|x|\cdot\frac{n+1}{n+2} \\
&= |x|\cdot 1
\end{aligned}$$

So ratio test guarantees convergence when $|x| < 1$, $R = 1$ then. We don't need to check endpoints since it only asks for the radius.

Example 11.9.2

Find a power series representation for $\arctan(x)$. Find its radius of convergence.

Solution

Again, we have to figure out how to relate this function to something like $\frac{1}{1-u}$. We know $\int \frac{1}{1+x^2}\,\mathrm{d}x = \arctan(x) + C$, so

$$\begin{aligned}
\arctan(x) &= \int \frac{1}{1+x^2}\,\mathrm{d}x \\
&= \int \frac{1}{1-(-x^2)}\,\mathrm{d}x \\
&= \int \sum_{n=0}^{\infty} (-x^2)^n\,\mathrm{d}x \\
&= \int \sum_{n=0}^{\infty} (-1)^n (x^{2n})\,\mathrm{d}x \\
&= \sum_{n=0}^{\infty} (-1)^n \int x^{2n}\,\mathrm{d}x \\
&= C + \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}
\end{aligned}$$

Now for finding C, we know $\arctan(0) = 0$, so

$$\begin{aligned}
0 &= C + \sum_{n=0}^{\infty} (-1)^n \frac{0^{2n+1}}{2n+1} \\
&= C
\end{aligned}$$

so $C = 0$. Then, a power series representation for $\arctan(x)$ is $\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}$.

Again, we can use ratio test to find the radius of convergence

$$\begin{aligned}
L &= \lim_{n\to\infty} \left| \frac{x^{2(n+1)+1}}{2(n+1)+1} \cdot \frac{2n+1}{x^{2n+1}} \right| \\
&= \lim_{n\to\infty} \left| \frac{x^{2n+3}}{2n+3} \cdot \frac{2n+1}{x^{2n+1}} \right| \\
&= \lim_{n\to\infty} |x^2| \frac{2n+1}{2n+3} \\
&= |x^2|
\end{aligned}$$

So the ratio test guarantees convergence when $L = |x| < 1$. Which is the same as $|x| < 1$ so $R = 1$.

Recall: we have 2 methods of estimating error when approximating series using finite sums.

Theorem 11.9.1 **Estimating Series' Error**

For $R_n = |S - S_n|$ where S is the series value and S_n is the nth partial sum we have

1. If $a_n = f(n)$, where f is a continuous, positive, eventually decreasing function and $\sum a_n$ converges, then the remainder R_n satisfies
$$\int_{n+1}^{\infty} f(x)\,\mathrm{d}x \leq R_n \leq \int_{n}^{\infty} f(x)\,\mathrm{d}x.$$
2. If $a_n = (-1)^{n+1} b_n$, $b_n \geq b_{n+1}$ for $n \geq N$, and $\lim_{n\to\infty} b_n = 0$, then
$$R_n \leq b_{n+1}.$$

Example 11.9.3

Find a power series representation of $\dfrac{1}{1+x^5}$

Solution

To find the power series representation we have to connect the function to something like $\frac{1}{1-u}$... $\frac{1}{1+x^5} = \frac{1}{1-(-x^5)}$ so

$$\begin{aligned}\frac{1}{1+x^5} &= \frac{1}{1-(-x^5)} \\ &= \sum_{n=0}^{\infty} (-x^5)^n \\ &= \sum_{n=0}^{\infty} (-1)^n x^{5n}\end{aligned}$$

Example 11.9.4

Using your answer to the previous example, approximate $\displaystyle\int_0^{0.3} \frac{1}{1+x^5}\,\mathrm{d}x$ correct to within 10^{-5}.

Solution

Since $\frac{1}{1+x^5} = \sum_{n=0}^{\infty} (-1)^n x^{5n}$, the integral

$$\begin{aligned}\int \frac{1}{1+x^5}\,\mathrm{d}x &= \sum_{n=0}^{\infty} (-1)^n \int x^{5n}\,\mathrm{d}x \\ &= C + \sum_{n=0}^{\infty} (-1)^n \frac{x^{5n+1}}{5n+1}\end{aligned}$$

Since we are approximating a definite integral, C does not matter, we can use $C = 0$.

$$\int_0^{0.3} \frac{1}{1+x^5}\,\mathrm{d}x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{5n+1}}{5n+1}\,\Big|_{x=0}^{0.3}$$
$$= \sum_{n=0}^{\infty} (-1)^n \frac{(0.3)^{5n+1}}{5n+1}$$

Notice that $\sum_{n=0}^{\infty} (-1)^n \frac{(0.3)^{5n+1}}{5n+1}$ is an alternating series with $b_n = \frac{(0.3)^{5n+1}}{5n+1} > 0$. (Exercise show that b_n is decreasing and $\lim_{n\to\infty} b_n = 0$) Then, but the alternating series estimation theorem

$$R_n < b_{n+1} < 10^{-5}$$

Observe that

$$b_{n+1} = \frac{(0.3)^{5(n+1)+1}}{5(n+1)+1} \approx \frac{0.3^{5n+6}}{5n+6}$$

and we can use a calculator or excel to find that $n = 2$ is the first n value to have $b_n + 1 < 10^{-5}$.

So we can approximate the series (and hence the integral) with only 2 terms to get within 10^{-5} of the correct answer!

$$\int_0^{0.3} \frac{1}{1+x^5}\,\mathrm{d}x \approx \sum_{n=0}^{2} (-1)^n \frac{(0.3)^{5n+1}}{5n+1} = \left(\frac{3}{10}\right) - \frac{1}{6}\left(\frac{3}{10}\right)^6$$

11.9b Section Summary:

- We continued to use $\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$ to write power series of other functions.
- We discussed the error in approximating an integral using a series. (Note you won't be tested on error approximations, but they are actually the most important results you get from this class)